**Tasks to Execute**

1. Execute all scripts from the **devops-cloud-documents-** channel.

**Bash Script Tasks**

1. Create a bash script to **check if a directory is available or not**.
2. Create a bash script to **create multiple files**.
3. Create a bash script to **take a backup of a directory**.
4. Create a bash script to **install Nginx on an EC2 server**.
5. Create a bash script to **install Apache Tomcat on an EC2 server**.
6. Create a bash script to **check if the Nginx service is running**, if not running then script should start the service.
7. Create a bash script for a **calculator**.
8. Create a bash script to **check if a directory exists**, if not then create a directory.
9. Create a bash script to **delete the last 3 lines of a file**.
10. Bash script to monitor cpu and if it is more than 80% then send email notification.
11. Bash script to monitor disk space and if it is more than 80% then send email notification.
12. Bash script to monitor memory and if it is more than 80% then send email notification.

I have created all 12 task in a bash script.

**1. Directory Existence Check**

Checked whether /root/sharikh exists and printed a result.

**Commands used:** if, test operator -e, echo.

**2. Create and List Multiple Files**

Created multiple files in /root/sharikh and listed them.

**Commands used:** touch, ls; brace expansion: {a,b,c,1,2,3,4,5}.

**3. Directory Backup**

Copied the directory and its contents to a backup location.

**Commands used:** cp -r, ls.

**4. Install and Start Nginx**

Attempted to install Nginx, check its service state, and start it.

**Commands used:** yum install nginx -y, systemctl status, grep, systemctl start, $? (last exit status).

**5. Install Java and Set Up Tomcat**

Updated packages, installed Java, downloaded and extracted Tomcat, then attempted to start it.

**Commands used:** yum update -y, yum install java -y, mkdir, wget -P, tar -xvzf, startup.sh, netstat -na, grep.

**6. Check and Start Nginx Again**

Repeated the Nginx status check and start attempt.

**Commands used:** systemctl status, grep inactive, systemctl start, command substitution $(...), echo.

**7. Bash Arithmetic Calculator**

Read two numbers and displayed addition, subtraction, multiplication, and division.

**Commands used:** read, echo, arithmetic expansion $((...)).

**8. Create a Directory if Missing**

Checked for /root/sharikh and created it only if it did not exist.

**Commands used:** if, -d, mkdir -p, variables ($DIR), echo.

**9. Delete the Last Three Lines of a File**

Asked for a file name, checked that it was a regular file, and removed the last three lines.

**Commands used:** read -p, if, -f, sed -i '$d', echo.

**10. CPU Usage Monitoring and Email Alert**

Estimated CPU usage and attempted to email an alert when it exceeded 80%.

**Commands used:** top -bn1, grep, awk, cut, variables, arithmetic expansion, if, hostname, mail.

**11. Disk Usage Monitoring and Email Alert**

Checked filesystem usage and attempted email alerts for mount points above 80%.

**Commands used:** df -P, awk, pipe (|), while read -r, parameter expansion ${USAGE%\%}, if, hostname, mail.

**12. Memory Usage Monitoring and Email Alert**

Read total and available memory, calculated usage, and attempted an alert above 80%.

**Commands used:** free -m, awk, variables, arithmetic expansion, if, hostname, mail, echo

```
#!/bin/bash
#owner = Sharikh khan
#modefied = 16th of oct. 2026

#1
echo "######### CHECK IF DIRECTORY IS AVAILABLE OR NOT #########"
echo
if [ -e /root/sharikh ]

then
        echo "directory exist"

else
        echo "directory does not exist"

fi
echo
echo


#2
echo  "######### CREATING MULTIPLE FILES ##########"
echo

touch /root/sharikh/{a,b,c,1,2,3,4,5}
echo
echo
echo "######### LIST THE FILES IN SHARIKH DIRECTORY ###########"
echo
ls /root/sharikh
echo
echo

#3
echo "####### TAKE BACKUP OF DIRECTORY SHARIKH #########"
echo
cp -r /root/sharikh /root/scripts/sharikh_backup
echo
ls /root/scripts
#4
echo "################################### INSTALLING NGINX#########################################################"
echo
```

```
#4
echo "################################### INSTALLING NGINX#########################################################"
echo
echo
yum install nginx -y > /dev/null

if  [ $? -eq 0 ]
then

echo "###################################### NGINX IS SUCCESFULLY INSTALLED ##########################################"
fi
echo
echo
echo "###################################### CHECK THE STATUS OF NGINX ###################################"
echo
echo
status=$(systemctl status nginx | grep inactive)
start=$(systemctl start nginx)
echo
echo $status

if [ $? -eq 0 ]
then
   $start
else
echo "################################################nginx is unable to start#######################################"
fi
echo
echo $start
echo "############################################ NGINX IS STARTED ########################################"


#5
echo "######################################## INSTALLING TOMCAT  ######################################"
echo
yum update -y
yum install java -y

mkdir /root/server1
wget -P /root/server1 https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.61/bin/apache-tomcat-10.1.61.tar.gz
tar -xvzf /root/server1/apache-tomcat-10.1.61.tar.gz
```

```
echo
echo " ########STARTING TOMCAT SERVIS USING ./STARTUP.SH ################################"
/root/scripts/apache-tomcat-10.1.61/bin/startup.sh
if [ $? -eq 0 ]
then
        echo "################################## TOMCAT HAS STARTED ############################"
echo
else
        echo "################################## UNABLE TO START TOMCAT ########################"
fi
echo
netstat -na | grep 8080

#6
echo "######################################## CHECK THE STATUS OF NGINX ###################################"
echo
echo
status=$(systemctl status nginx | grep inactive)
start=$(systemctl start nginx)
echo
echo $status

if [ $? -eq 0 ]
then
echo $start
else
echo "##############################################nginx is unable to start#########################################"
fi
echo $status

echo nginx is active


#7

echo "Enter first number:"
read a

echo "Enter second number:"
read b
```

```
echo "Enter second number:"
read b

echo "Addition: $((a + b))"
echo "Subtraction: $((a - b))"
echo "Multiplication: $((a * b))"
echo "Division: $((a / b))"


#8
echo
echo  "#################### CHECK IF DIRECTORY IS AVAILABLE IF NOT CREATE NEW ONE ###################################"
echo
DIR="/root/sharikh"

if [ -d "$DIR" ]; then
    echo "Directory already exists"
else
    mkdir -p "$DIR"
    echo "Directory created successfully"
fi

#9
read -p "Enter file name: " file

if [ -f "$file" ]; then
    sed -i '$d' "$file"
    sed -i '$d' "$file"
    sed -i '$d' "$file"

    echo "Last 3 lines deleted successfully"
else
    echo "Error: File does not exist"
fi

#10
echo "############################### monitor cpu and if it is more than 80% then send email ############################"
echo
THRESHOLD=80
EMAIL="admin@example.com"
```

```
EMAIL="admin@example.com"

# Get CPU usage percentage
CPU_IDLE=$(top -bn1 | grep "Cpu(s)" | awk '{print $8}' | cut -d. -f1)
CPU_USAGE=$((100 - CPU_IDLE))

echo "Current CPU Usage: ${CPU_USAGE}%"

# Check CPU usage
if [ "$CPU_USAGE" -gt "$THRESHOLD" ]; then
    echo "WARNING: CPU usage is above ${THRESHOLD}%!"

    MESSAGE="Alert: CPU usage is ${CPU_USAGE}% on $(hostname)"

    echo "$MESSAGE" | mail -s "High CPU Usage Alert" "$EMAIL"

    echo "Email notification sent."
else
    echo "CPU usage is normal."
fi

#11
echo "############################################## monitor disk space and if it is more than 80% then send email ##############################"

THRESHOLD=80
EMAIL="admin@example.com"


df -P -x tmpfs -x devtmpfs | awk 'NR>1 {print $1, $5, $6}' |
while read -r DEVICE USAGE MOUNTPOINT
do
    USAGE=${USAGE%\%}

    if [ "$USAGE" -gt "$THRESHOLD" ]; then

        echo "WARNING: Disk usage is above ${THRESHOLD}%!"
        echo "Device: $DEVICE"
        echo "Mount Point: $MOUNTPOINT"
        echo "Usage: ${USAGE}%"

        MESSAGE="Disk Alert on $(hostname):
                                                                                                    194,2         75%
```

```
Device: $DEVICE
Disk Usage: ${USAGE}%
Threshold: ${THRESHOLD}%"

        echo "$MESSAGE" | mail -s "Disk Space Alert" "$EMAIL"

        echo "Email notification attempted."
        echo
    else
        echo "OK: $MOUNTPOINT usage is ${USAGE}%"
    fi
done

#12
echo ################################## monitor memory and if it is more than 80% then send email ######################
echo
THRESHOLD=80
EMAIL="admin@example.com"

echo "######## CHECKING MEMORY USAGE ########"

# Get total and available memory in MB
TOTAL=$(free -m | awk '/^Mem:/ {print $2}')
AVAILABLE=$(free -m | awk '/^Mem:/ {print $7}')

# Calculate memory usage percentage
USED=$((TOTAL - AVAILABLE))
USAGE=$((USED * 100 / TOTAL))

echo "Total Memory: ${TOTAL} MB"
echo "Used Memory: ${USED} MB"
echo "Memory Usage: ${USAGE}%"

# Check memory usage
if [ "$USAGE" -gt "$THRESHOLD" ]; then

    echo "WARNING: Memory usage is above ${THRESHOLD}%!"

    MESSAGE="Memory Alert on $(hostname):
Memory Usage: ${USAGE}%
Total Memory: ${TOTAL} MB
```

```
done

#12
echo ################################## monitor memory and if it is more than 80% then send email #######
echo
THRESHOLD=80
EMAIL="admin@example.com"

echo "######## CHECKING MEMORY USAGE ########"

# Get total and available memory in MB
TOTAL=$(free -m | awk '/^Mem:/ {print $2}')
AVAILABLE=$(free -m | awk '/^Mem:/ {print $7}')

# Calculate memory usage percentage
USED=$((TOTAL - AVAILABLE))
USAGE=$((USED * 100 / TOTAL))

echo "Total Memory: ${TOTAL} MB"
echo "Used Memory: ${USED} MB"
echo "Memory Usage: ${USAGE}%"

# Check memory usage
if [ "$USAGE" -gt "$THRESHOLD" ]; then

    echo "WARNING: Memory usage is above ${THRESHOLD}%!"

    MESSAGE="Memory Alert on $(hostname):
Memory Usage: ${USAGE}%
Total Memory: ${TOTAL} MB
Used Memory: ${USED} MB
Threshold: ${THRESHOLD}%"

    echo "$MESSAGE" | mail -s "High Memory Usage Alert" "$EMAIL"

    echo "Email notification attempted."

else
    echo "Memory usage is normal."
fi
```

```
[root@ip-172-31-26-124 scripts]# ./bash-task.bash
######### CHECK IF DIRECTORY IS AVAILABLE OR NOT #########

directory exist


######## CREATING MULTIPLE FILES ###########



######### LIST THE FILES IN SHARIKH DIRECTORY ############

1  2  3  4  5  a  b  c


####### TAKE BACKUP OF DIRECTORY SHARIKH #########


 1  '\'   bash-task.bash   sharikh_backup   test.bash   test.txt
###################################### INSTALLING NGINX#############################################################


######################################## NGINX IS SUCCESFULLY INSTALLED ##############################################


######################################### CHECK THE STATUS OF NGINX ###################################



Active: inactive (dead)


############################################### NGINX IS STARTED ##############################################
########################################### INSTALLING TOMCAT  ################################################

Last metadata expiration check: 4:07:40 ago on Fri Oct 16 10:11:05 2026.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 4:07:42 ago on Fri Oct 16 10:11:05 2026.
Package java-27-amazon-corretto-1:27.0.0+36-1.amzn2023.1.x86_64 is already installed.
Dependencies resolved.
```

```
--2026-10-16 14:18:47--  https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.61/bin/apache-tomcat-10.1.61.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14488199 (14M) [application/x-gzip]
Saving to: '/root/server1/apache-tomcat-10.1.61.tar.gz.2'

apache-tomcat-10.1.61.tar.gz.2          100%[=============================================================================>]  13.82M  --.-KB/s    in 0.04s

2026-10-16 14:18:48 (329 MB/s) - '/root/server1/apache-tomcat-10.1.61.tar.gz.2' saved [14488199/14488199]

apache-tomcat-10.1.61/conf/
apache-tomcat-10.1.61/conf/catalina.policy
apache-tomcat-10.1.61/conf/catalina.properties
apache-tomcat-10.1.61/conf/context.xml
apache-tomcat-10.1.61/conf/jaspic-providers.xml
apache-tomcat-10.1.61/conf/jaspic-providers.xsd
apache-tomcat-10.1.61/conf/logging.properties
apache-tomcat-10.1.61/conf/server.xml
apache-tomcat-10.1.61/conf/tomcat-users.xml
apache-tomcat-10.1.61/conf/tomcat-users.xsd
apache-tomcat-10.1.61/conf/web.xml
apache-tomcat-10.1.61/bin/
apache-tomcat-10.1.61/lib/
apache-tomcat-10.1.61/logs/
apache-tomcat-10.1.61/temp/
apache-tomcat-10.1.61/webapps/
apache-tomcat-10.1.61/webapps/ROOT/
apache-tomcat-10.1.61/webapps/ROOT/WEB-INF/
apache-tomcat-10.1.61/webapps/docs/
apache-tomcat-10.1.61/webapps/docs/META-INF/
apache-tomcat-10.1.61/webapps/docs/WEB-INF/
apache-tomcat-10.1.61/webapps/docs/WEB-INF/jsp/
apache-tomcat-10.1.61/webapps/docs/annotationapi/
apache-tomcat-10.1.61/webapps/docs/api/
apache-tomcat-10.1.61/webapps/docs/appdev/
apache-tomcat-10.1.61/webapps/docs/appdev/sample/
apache-tomcat-10.1.61/webapps/docs/appdev/sample/docs/
apache-tomcat-10.1.61/webapps/docs/appdev/sample/src/
apache-tomcat-10.1.61/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-10.1.61/webapps/docs/appdev/sample/web/
apache-tomcat-10.1.61/webapps/docs/appdev/sample/web/WEB-INF/
pport MobaXterm by subscribing to the professional edition here:  https://mobaxterm.mobatek.net
```

```
apache-tomcat-10.1.61/webapps/manager/css/manager.css
apache-tomcat-10.1.61/webapps/manager/images/asf-logo.svg
apache-tomcat-10.1.61/webapps/manager/images/favicon.ico
apache-tomcat-10.1.61/webapps/manager/images/tomcat.svg
apache-tomcat-10.1.61/webapps/manager/index.jsp
apache-tomcat-10.1.61/webapps/manager/status.xsd
apache-tomcat-10.1.61/webapps/manager/xform.xsl
apache-tomcat-10.1.61/bin/catalina.sh
apache-tomcat-10.1.61/bin/ciphers.sh
apache-tomcat-10.1.61/bin/configtest.sh
apache-tomcat-10.1.61/bin/daemon.sh
apache-tomcat-10.1.61/bin/digest.sh
apache-tomcat-10.1.61/bin/makebase.sh
apache-tomcat-10.1.61/bin/migrate.sh
apache-tomcat-10.1.61/bin/setclasspath.sh
apache-tomcat-10.1.61/bin/shutdown.sh
apache-tomcat-10.1.61/bin/startup.sh
apache-tomcat-10.1.61/bin/tool-wrapper.sh
apache-tomcat-10.1.61/bin/version.sh

 #######STARTING TOMCAT SERVIS USING ./STARTUP.SH ##############################
Using CATALINA_BASE:   /root/scripts/apache-tomcat-10.1.61
Using CATALINA_HOME:   /root/scripts/apache-tomcat-10.1.61
Using CATALINA_TMPDIR: /root/scripts/apache-tomcat-10.1.61/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /root/scripts/apache-tomcat-10.1.61/bin/bootstrap.jar:/root/scripts/apache-tomcat-10.1.61/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
############################### TOMCAT HAS STARTED ############################


################################### CHECK THE STATUS OF NGINX ###################################




nginx is active
Enter first number:
1
Enter second number:
```

```
 ######################################## CHECK THE STATUS OF NGINX ####################################




 nginx is active
 Enter first number:
 1
 Enter second number:
 5
 Addition: 6
 Subtraction: -4
 Multiplication: 5
 Division: 0

 ################### CHECK IF DIRECTORY IS AVAILABLE IF NOT CREATE NEW ONE ####################################

 Directory already exists
 Enter file name: test.txt
 Last 3 lines deleted successfully
 ############################## monitor cpu and if it is more than 80% then send email ############################

 Current CPU Usage: 7%
 CPU usage is normal.
 ############################################## monitor disk space and if it is more than 80% then send email ########################
 OK: /sys/firmware/efi/efivars usage is 3%
 OK: / usage is 26%
 OK: /boot/efi usage is 13%

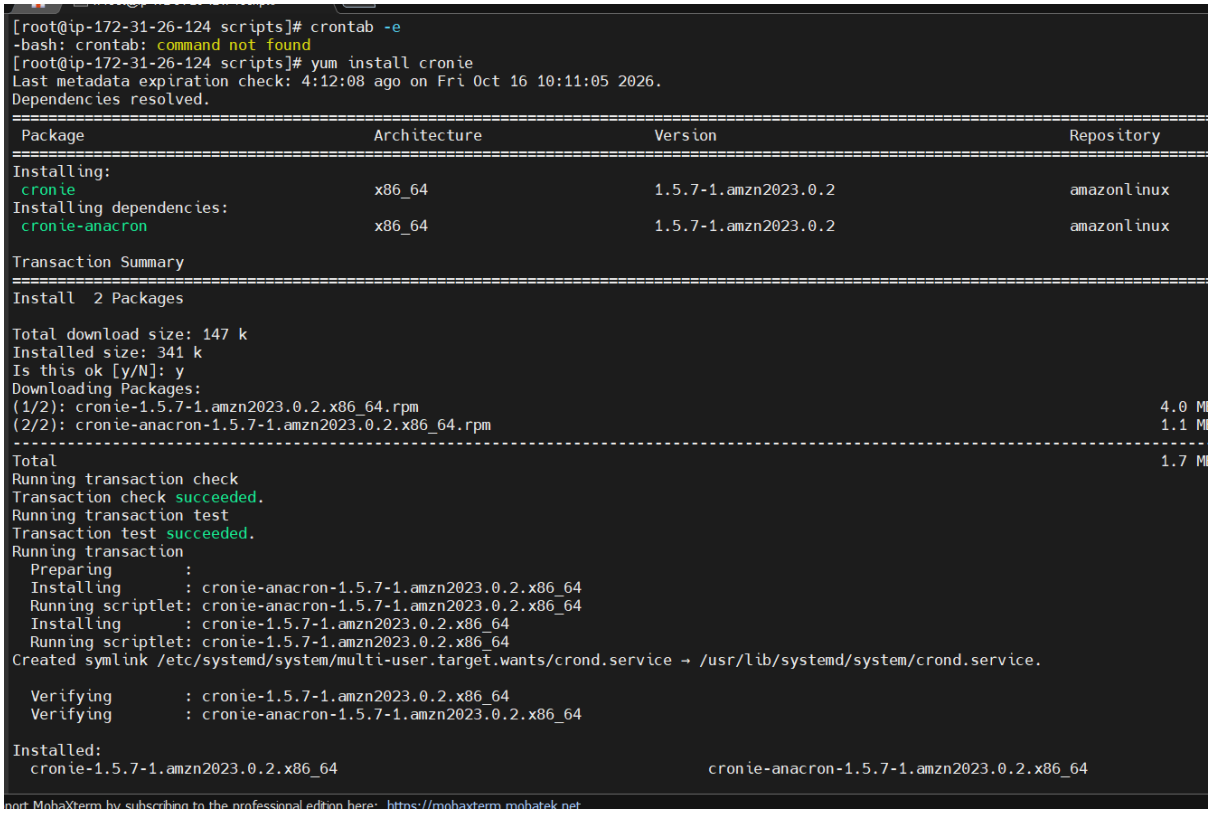

 ######## CHECKING MEMORY USAGE ########
 Total Memory: 913 MB
 Used Memory: 424 MB
 Memory Usage: 46%
 Memory usage is normal.
 [root@ip-172-31-26-124 scripts]#
```

**Exercise: Crontab Entries**

1. April 5th Midnight
2. 5th of Every November, January, June **if it is a Thursday**
3. At **05 and 27th minutes** of **9, 10, 11** hours every day
4. **34 minutes** of **9th hour** on **15th August**
5. Every midnight
6. Every Weekend (**Saturday night 11:59 PM**)
7. After every reboot

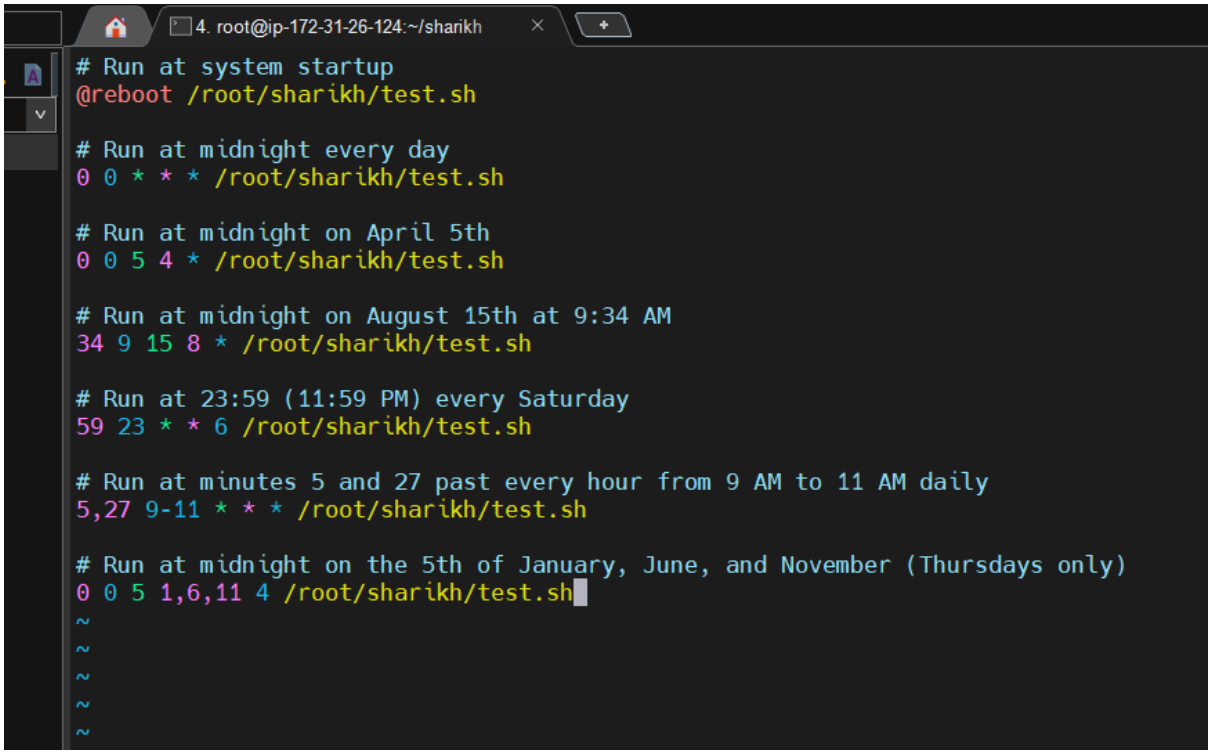
Install crontab using

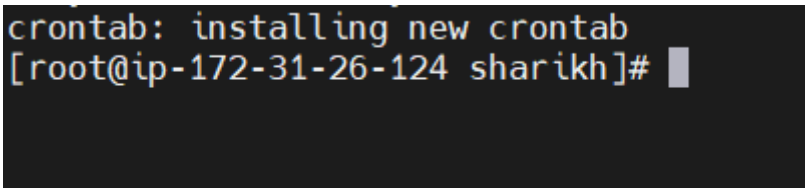
$ Yum install cronie




After that add crontab using

Crontab -e




Install Git using

$ Yum install git

```
[root@ip-172-31-26-124 scripts]# yum install git
Last metadata expiration check: 4:45:58 ago on Fri Oct 16 10:11:05 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture            Version
================================================================================
Installing:
 git                     x86_64                  2.50.1-1.amzn2023.0.1
Installing dependencies:
 git-core                x86_64                  2.50.1-1.amzn2023.0.1
 git-core-doc            noarch                  2.50.1-1.amzn2023.0.1
 perl-Error              noarch                  1:0.17030-2.amzn2023.0.1
 perl-File-Find          noarch                  1.37-477.amzn2023.0.10
 perl-Git                noarch                  2.50.1-1.amzn2023.0.1
 perl-TermReadKey        x86_64                  2.38-9.amzn2023.0.3
 perl-lib                x86_64                  0.65-477.amzn2023.0.10

Transaction Summary
================================================================================
Install  8 Packages

Total download size: 7.9 M
Installed size: 41 M
Is this ok [y/N]: y
Downloading Packages:
(1/8): git-2.50.1-1.amzn2023.0.1.x86_64.rpm
(2/8): git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm
(3/8): perl-Error-0.17030-2.amzn2023.0.1.noarch.rpm
(4/8): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm
(5/8): perl-File-Find-1.37-477.amzn2023.0.10.noarch.rpm
(6/8): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm
(7/8): perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64.rpm
(8/8): perl-lib-0.65-477.amzn2023.0.10.x86_64.rpm
--------------------------------------------------------------------------------
Total
Running transaction check
Transaction check succeeded.
Running transaction test
```

Initialize git using

$git init

```
[root@ip-172-31-26-124 scripts]# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /root/scripts/.git/
[root@ip-172-31-26-124 scripts]#
```

Add my file to git to track using

Git add filename

Git status

```
 1  '\'   apache-tomcat-10.1.61   bash-task.bash   sharikh_backup   test.bash   test.
[root@ip-172-31-26-124 scripts]# git add bash-task.bash
[root@ip-172-31-26-124 scripts]# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   bash-task.bash

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        1
        "\\"
        apache-tomcat-10.1.61/
        sharikh_backup/
        test.bash
        test.txt

[root@ip-172-31-26-124 scripts]#
```

Commit git filr using

$ git commit -m “comment”

```
[root@ip-172-31-26-124 scripts]# git commit -m "adding bash script"
[master (root-commit) 7ef83e0] adding bash script
 Committer: root <root@ip-172-31-26-124.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 252 insertions(+)
 create mode 100755 bash-task.bash
[root@ip-172-31-26-124 scripts]#
```

Add remote git origin

```
After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

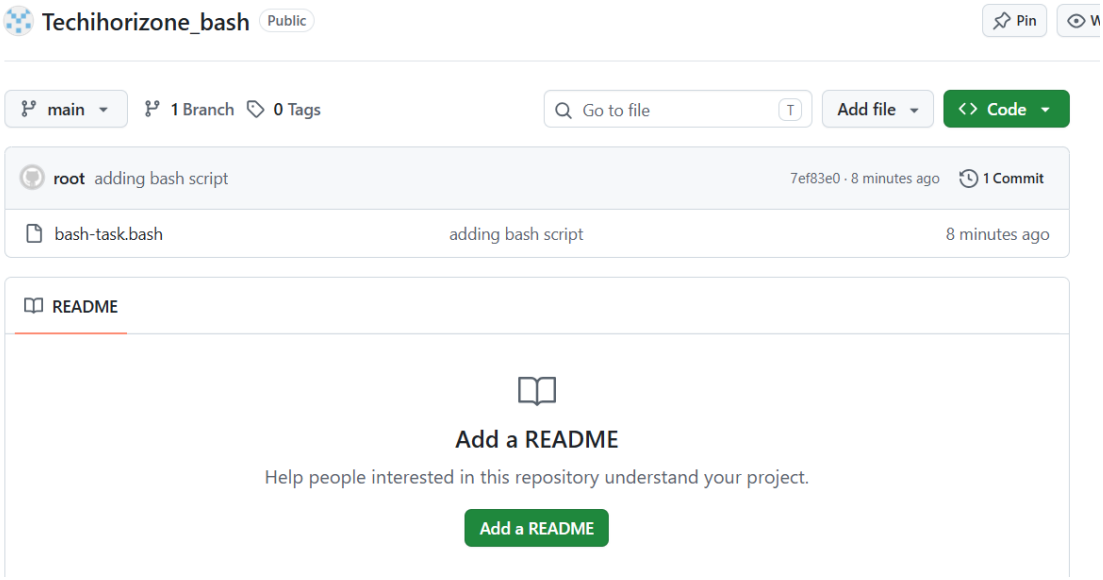
 1 file changed, 252 insertions(+)
 create mode 100755 bash-task.bash
[root@ip-172-31-26-124 scripts]# git remote add origin https://github.com/khansharikh/Techihorizone_bash.git
[root@ip-172-31-26-124 scripts]#
```

git branch -M main

git push -u origin main

get token from developer setting for authentication./

```
[root@ip-172-31-26-124 scripts]# git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/khansharikh/Techihorizone_bash.git'
[root@ip-172-31-26-124 scripts]# git branch -M main
[root@ip-172-31-26-124 scripts]# git push -u origin main
^M
[root@ip-172-31-26-124 scripts]# git push -u origin main
Username for 'https://github.com': khansharikh
Password for 'https://khansharikh@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/khansharikh/Techihorizone_bash.git/'
[root@ip-172-31-26-124 scripts]# git push -u origin main
Username for 'https://github.com': khansharikh
Password for 'https://khansharikh@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 1.92 KiB | 1.92 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/khansharikh/Techihorizone_bash.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
[root@ip-172-31-26-124 scripts]#
```




https://github.com/khansharikh/Techihorizone_bash

above the link for my repo.